

1 Poisson Distribution

Poisson Distribution

$X \sim \text{Poi}(\lambda)$ if $p(x) = \frac{\lambda^x}{x!} e^{-\lambda}$ for $x = 0, 1, 2, \dots$
 $E[X] = \lambda, \quad \text{Var}(X) = \lambda, \quad E[X^2] = \lambda^2 + \lambda$
 $M_X(t) = E[e^{tX}] = e^{\lambda(e^t - 1)}$
Ind. X and $Y \sim \text{Poi}(\mu) \implies X + Y \sim \text{Poi}(\lambda + \mu)$
 $Z|X \sim \text{Bin}(X, r) \implies Z \sim \text{Poi}(r\lambda)$

Exponential Distribution

$X \sim \text{Exp}(\lambda)$ if
 $f(x) = \lambda e^{-\lambda x}, F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$.
 $E[X] = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}$.
 $M_X(t) = E[e^{tX}] = \frac{\lambda}{\lambda - t}, \quad t < \lambda$.

1.1 Poisson Process

A **Poisson process with rate** $\lambda > 0$ is an integer-valued stochastic process $\{X(t), t \geq 0\}$ s.t.

- for any time points $t_0 = 0 < t_1 < t_2 < \dots < t_n$, the increments $X(t_1) - X(t_0), X(t_2) - X(t_1), \dots, X(t_n) - X(t_{n-1})$ are independent R.V.s
- for $s \geq 0, t > 0, X(s+t) - X(s) \sim \text{Poi}(\lambda t)$
- $X(0) = 0$

At any time point $t > 0$:

$$X(t) \sim \text{Poi}(\lambda t)$$
$$E[X(t)] = \lambda t, \quad \text{Var}(X(t)) = \lambda t$$
$$M_{X(t)}(u) = e^{\lambda t(e^u - 1)}$$

Rare Events

A **rare event** is an event s.t. the total occurrences of the event ϵ_h in $h \rightarrow 0$ units of time is given by:

- $P(\epsilon_h = 0) = 1 - \lambda h + o(h)$
- $P(\epsilon_h = 1) = \lambda h + o(h)$
- $P(\epsilon_h \geq 2) = o(h)$

For independent Bernoulli R.V.s $\epsilon_1, \epsilon_2, \dots$ with $P(\epsilon_i) = p_i$, and their sum $S_n = \epsilon_1 + \epsilon_2 + \dots + \epsilon_n$,

$$\left| P(S_n = k) - \frac{\lambda^k e^{-\lambda}}{k!} \right| \leq \sum_{i=1}^n p_i^2.$$

An R.V. $X((s, t])$ counting the **number of occurrences of a rare event** in the time interval $(s, t]$ is a **non-homogeneous Poisson process** with rate $\lambda(t) > 0$ if:

- for any time points $t_0 = 0 < t_1 < t_2 < \dots < t_n$, the process increments $X((t_0, t_1]), X((t_1, t_2]), \dots, X((t_{n-1}, t_n])$ are independent R.V.s
- $\exists$ a positive function $\lambda(t) > 0$ for which the prob. of at least one event occurrence in a small time interval $h \rightarrow 0$ is given by:
 - $P(X((t, t+h]) = 1) = \lambda(t)h + o(h)$
 - $P(X((t, t+h]) \geq 2) = o(h)$

Each increment $X((s, t])$ of a non-homogeneous Poisson process follows

$$X((s, t]) \sim \text{Poi}\left(\int_s^t \lambda(u) du\right).$$

If $\lambda(t)$ itself is a random process, then $X((s, t])$ is called a **doubly stochastic Poisson process** or a **Cox process**.

A **Poisson point process** is a special case of the non-homogeneous Poisson process with rate $\lambda(t) = \lambda$ for all $t \geq 0$.

Arrival Times

The **waiting time/arrival time** W_n is the time of occurrence of the n -th event, supposing $W_0 = 0$. $W_n \sim \text{Gamma}(n, \lambda)$.

$$P(W_n > t) = P(X(t) < n) = \sum_{k=0}^{n-1} \frac{(\lambda t)^k}{k!} e^{-\lambda t}.$$

$$f_{W_n}(t) = \frac{\lambda^n t^{n-1}}{(n-1)!} e^{-\lambda t}, \quad n \in \mathbb{N}^*, t \geq 0.$$

The **sojourn time** S_n for the n -th occurrence of a Poisson process is the time interval between the $(n-1)$ -th and n -th occurrence.

$$S_n = W_{n+1} - W_n, \quad S_n \sim \text{exp}(\lambda).$$

Suppose $\{S_n : n \geq 0\}$ of i.i.d. $\text{exp}(\lambda)$ R.V.s. A counting process defined by the arrival time of the n -th event

$$W_n = S_0 + S_1 + \dots + S_{n-1}$$

is a **Poisson process with rate** λ .

1.2 Properties

Short-term Performance

- $X(t) \sim \text{Poi}(\lambda t) \quad t \geq 0$
- $X(t+s) | [X(t) = k] \sim k + \text{Poi}(\lambda s)$
- $P(X(t) = i | [X(t+s) = k])$ is given by Bayes' theorem:
$$\frac{P(X(t) = i) P(X(t+s) - X(t) = k - i)}{P(X(t+s) = k)}$$

Conditional Distribution

Given $X(t) = n$, the arrival times $W_1, W_2, \dots, W_n$ have the same distribution as the **order statistics** of n i.i.d. $\text{Uniform}(0, t)$ R.V.s. That is, for $x \in (0, t), k \in \mathbb{N}_{\leq n}^*$:

$$f_k(x) = \frac{n!}{(k-1)!(n-k)!} \left(\frac{x}{t}\right)^{k-1} \left(1 - \frac{x}{t}\right)^{n-k} \frac{1}{t}.$$

Note: The Poisson counting process is characterised by the waiting times, with a fixed jumping pattern, no revisits, no absorbing states, and no stationary distribution.

1.3 Multiple Poisson Processes

Merging Poisson Processes: Consider independent poisson processes $\{N_1(t), t \geq 0\}, \dots, \{N_m(t), t \geq 0\}$ with rates $\lambda_1, \dots, \lambda_m$ respectively. $N(t) := \sum_{i=1}^m N_i(t)$. Then $\{N(t), t \geq 0\}$ is a P.P. with rate $\lambda = \sum_{i=1}^m \lambda_i$.

Splitting Poisson Processes: Consider a Poisson process $\{N(t), t \geq 0\}$ with rate λ . Each event occurrence is independently classified as a type i event with prob. r_i for $i = 1, 2, \dots, m$, where $\sum_{i=1}^m r_i = 1$.

For counting processes of type i events $N_i(t)$, each $\{N_i(t), t \geq 0\}$ is a P.P. with rate $r_i \lambda$ for $i = 1, 2, \dots, m$, and are independent P.P.s.

Waiting Time Comparison: Consider two independent Poisson processes $\{X(t), t \geq 0\}$ and $\{Y(t), t \geq 0\}$ with rates λ_1 and λ_2 respectively. Let $W_n^{(1)}$ be the waiting time of the n -th event of $X(t)$ and $W_m^{(2)}$ be the waiting time of the m -th event of $Y(t)$.

The probability $P(W_n^{(1)} < W_m^{(2)})$ is given by

$$\sum_{k=n}^{n+m-1} \binom{n+m-1}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n+m-1-k}.$$

Intuition: Consider the merged poisson process. Each event is from $X(t)$ with probability $\frac{\lambda_1}{\lambda_1 + \lambda_2}$. We consider bernoulli trials of obtaining n "heads" before m "tails".

1.4 Generalisations of Poisson Processes

Non-homogeneous Poisson Process: Consider a non-homogeneous Poisson process $\{X(t), t \geq 0\}$ with rate $\lambda(t) > 0$. The probability of at least one event occurrence in a small time interval $h \rightarrow 0$ is given by:

- $P(X((t, t+h]) = 1) = \lambda(t)h + o(h)$
- $P(X((t, t+h]) \geq 2) = o(h)$

Compound Poisson Process: Consider a Poisson process $\{N(t), t \geq 0\}$ with rate λ . Let $\{Y_n : n \in \mathbb{N}^*\}$ be i.i.d. R.V.s independent of $\{N(t), t \geq 0\}$. The **compound Poisson process** $\{X(t), t \geq 0\}$ is defined by

$$X(t) = \sum_{n=1}^{N(t)} Y_n.$$

Conditional (Mixed) Poisson Process: Consider a random variable Λ independent of a Poisson process $\{X(t), t \geq 0\}$ with rate Λ . The **conditional (mixed) Poisson process** $\{Y(t), t \geq 0\}$ is defined by

$$Y(t) = X(t) | [\Lambda].$$

Multi-dimensional Poisson Process: Consider a Poisson process $\{N(t), t \geq 0\}$ with rate λ . Let $\{Y_n : n \in \mathbb{N}^*\}$ be i.i.d. R.V.s independent of $\{N(t), t \geq 0\}$ and taking values in $\mathbb{R}^d$ for some $d \in \mathbb{N}^*$. The **multi-dimensional Poisson process** $\{X(t), t \geq 0\}$ is defined by

$$X(t) = \sum_{n=1}^{N(t)} Y_n.$$

2 Continuous Time Markov Chain

A **continuous time Markov chain** is a stochastic process $\{X(t), t \geq 0\}$ with a countable state space S s.t. for any time points $t_0 = 0 < t_1 < t_2 < \dots < t_n$ and any states $i_0, i_1, i_2, \dots, i_n$,

$$P(X(t_n) = i_n | X(t_{n-1}) = i_{n-1}, \dots, X(t_0) = i_0) = P(X(t_n) = i_n | X(t_{n-1}) = i_{n-1})$$